SCHOOL OF COMPUTING
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A Leap Towards World Class Education

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Name of the Experiment:

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Marks	Teacher's Signature with date
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Q1 Define an algorithm. List any 4 characteristics of an algorithm.

→ An algorithm is an finite sequence of well defined, step by step instruction used to solve a specific problem or perform a computation.

★ Characteristics.

1. Input: An algorithm takes 0 or more inputs
2. Output: It produces atleast one output
3. Definiteness: Each step is ~~close~~ clear, precise unambiguous.
4. finiteness: The algorithm terminals after a finite no of steps.

Q2: Define data structure and abstract data type (ADT). Give one example of each

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐

Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐

Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signatu

→ A data structure is a way of organising storing & managing data in memory so that it can be accessed & modified efficiently.

An abstract data type (ADT) is a logical description of a data type specifies the operations that can be performed on it without specifying how these operations are implemented.

Examples:
- Data structure : Array
- ADT : Stack (operations : push, Pop, peek)

Q3 Explain the process of converting a problem into a program. Illustrate the role of algorithms in the process.

→ 1) ~~Explain~~ Problem definition : Understand requirement and constraints.
2) Analysis : Identify inputs, outputs & processing
3) Algorithm des: Create step by step logic to solve the problem.
4) flowchart : Represent the algorithm visually or textually.
5) Coding : Convert algorithm into a program using a programming language.
6) Testing & debugging → check correction & remove error.

7. Documentation & maintainance : Improve & maintain the program.

Role of algorithm: acts as a blueprint of soln It corrects efficiency & makes coding easier & error free.

Q4. Explain linear & non-linear data structure with suitable examples.

→ Linear Data structure
data elements are arranged Sequentically
- Array
- Stack
- Queue
- linked list.

Non linear Data struct. Data elements are arranged in a hierarchical or interconnected manner.
Example :
- Tree
- Graph

Q5. Differentiate b/w algo & Prog. Analyse their role in SDLC.

Algorithm	Prog.
Step by step logic	written code
language independent	language dependant
Des level	Implementation level.
No syntax rule	follows syntax rule.

Q6. Examine the relationship among data struct & algorithm.

→ Relationship:
Data is raw information
Data structure organizes the data
Algorithm processes data using data struc.

- Justification:
The efficiency of an algorithm depends on the data structure used.
- Searching is an array → slower.
- Searching in a binary search tree.
- faster (O(logn)).

Conclusion:
Choosing the right data structure improves time & space efficiency of algorithms.